

Acuity Engine Monetary Impact Review

Executive summary: this platform should be evaluated as a hospital operations risk-reduction and workforce-efficiency tool. Its most credible financial impact comes from three areas:

- lower nursing turnover and staffing friction,
- fewer preventable hospital-acquired events,
- more consistent assignment decisions under interruption-heavy conditions.

1. Framing

This report presents **scenario-based financial modeling**, not guaranteed savings. The aim is to estimate the economic value that could plausibly be captured if a hospital uses the platform to improve assignment quality, reduce workforce strain, and prevent a subset of avoidable adverse events.

2. Source-Backed Cost Inputs

Cost category	Input used	Source basis
1% RN turnover change	\$289,000 per average hospital per year	2025 NSI figures cited by the American Hospital Association workforce scan
Replace one bedside RN	\$61,110 average	2025 NSI figures cited by AHA workforce scan / Becker summary
CLABSI	\$48,108 additional cost per case	AHRQ hospital-acquired condition meta-analysis
CAUTI	\$13,793 additional cost per case	AHRQ hospital-acquired condition meta-analysis
Inpatient falls	\$6,694 additional cost per case	AHRQ hospital-acquired condition meta-analysis
Hospital-acquired pressure injury	\$14,506 additional cost per case	AHRQ hospital-acquired condition meta-analysis
Severe pressure injury sensitivity	\$20,900 to \$151,700 per injury	AHRQ pressure injury training materials citing CMS-era estimates

3. Why This Platform Could Influence Those Costs

- More balanced assignments can reduce preventable workload concentration.
- Lower assignment volatility can reduce handoff complexity and hidden burden stacking.
- More consistent workload fairness may support retention and reduce burnout-related turnover pressure.
- Structured patient-risk visibility may improve prevention focus on falls, pressure injuries, CLABSI, and CAUTI.

Important note: these links are operational inferences, not proof of direct causality. The strongest evidence is that staffing strain and nurse workload affect burnout, dissatisfaction, and some patient safety outcomes; the platform's role is to improve assignment quality within that operating environment.

4. Modeled Annual Savings Scenarios

The following scenarios assume the platform contributes to a reduction in turnover and a modest reduction in preventable events over a year at a single hospital.

Scenario	Assumptions	Estimated annual savings
Conservative	1% RN turnover improvement 1 CLABSI prevented 3 CAUTIs prevented 5 falls prevented 4 pressure injuries prevented	\$438,503
Moderate	2% RN turnover improvement 2 CLABSIs prevented 6 CAUTIs prevented 10 falls prevented 8 pressure injuries prevented	\$877,006
Higher-impact	3% RN turnover improvement 4 CLABSIs prevented 10 CAUTIs prevented 15 falls prevented 12 pressure injuries prevented	\$1,402,735

5. Severe Pressure Injury Sensitivity

If the avoided pressure injuries are concentrated in severe Stage 3/4, unstageable, or DTI cases, financial effect may be materially higher.

Prevented severe HAPI cases	Low estimate	High estimate
3 cases	\$62,700	\$455,100
5 cases	\$104,500	\$758,500

Prevented severe HAPI cases	Low estimate	High estimate
10 cases	\$209,000	\$1,517,000

6. Workforce and Burnout Value

Recent workforce reporting cited by the American Hospital Association indicates that:

- average bedside RN replacement cost is about **\$61,110**,
- average hospital losses from nurse turnover are in the range of **\$3.9M to \$5.7M annually**,
- each 1% RN turnover change is worth about **\$289,000 per year** to the average hospital.

If the platform helps reduce unfair assignments, visible burden stacking, and preventable staffing dissatisfaction, even a small retention effect can matter financially.

7. Why This Is Hard To Do Manually

Assignment quality depends on balancing many simultaneous constraints under constant interruption. Manual assignment making on paper or in Excel depends on working memory and repeated mental recalculation. A structured balancing platform can re-run those comparisons consistently after each data change.

8. Key Caveats

- This report does **not** claim the platform alone directly prevents infections or injuries.
- Actual savings depend on adoption, workflow fit, baseline event rates, and quality of local implementation.
- Financial impact is best validated with pilot data: event trends, turnover trends, reassignment time, and user adoption.

9. Suggested ROI Framing For Hospitals

A reasonable framing is:

ROI case = turnover savings + avoided hospital-acquired-event costs + avoided operational friction from manual rework.

Even modest gains can become meaningful because turnover and preventable adverse events are each individually expensive.

10. Sources

- [AHRQ hospital-acquired condition cost estimates](#)
- [AHRQ pressure injury cost training materials](#)
- [AHRQ national declines in hospital-acquired conditions](#)

- [American Hospital Association workforce scan citing 2025 NSI figures](#)
- [Aiken et al. on nurse staffing, burnout, and dissatisfaction](#)
- [PubMed study on staffing and hospital-acquired pressure ulcers](#)
- [PubMed study on staffing and falls performance](#)

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